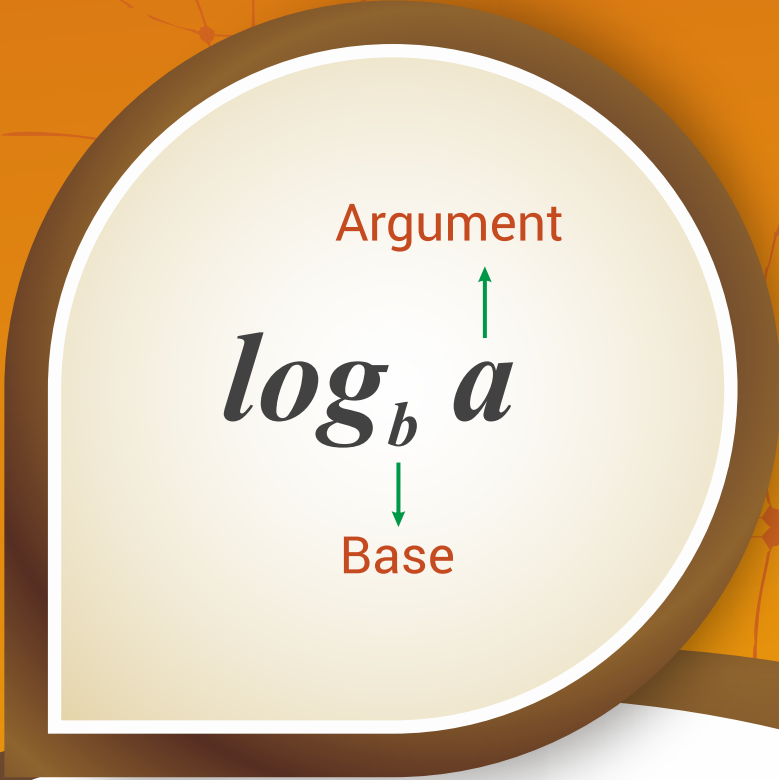


MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Argument

$$\log_b a$$

Base

The diagram shows the logarithm expression $\log_b a$ inside a light beige rounded rectangle. A green arrow points from the word 'Argument' above to the variable 'a'. Another green arrow points from the word 'Base' below to the subscript 'b'.

Exercise

Logarithm

(English Medium)

EXERCISE (O-1)

Straight Objective Type

1. If $a^4 \cdot b^5 = 1$ then the value of $\log_a(a^5 b^4)$ equals
 (A) $9/5$ (B) 4 (C) 5 (D) $8/5$ **LG0142**
2. The expression $\log_p \underbrace{\sqrt[p]{\sqrt[p]{\sqrt[p]{\dots \sqrt[p]{p}}}}}_{n \text{ radical sign}}$, where $p \geq 2, p \in \mathbb{N}; n \in \mathbb{N}$ when simplified is
 (A) independent of p (B) independent of p and of n
 (C) dependent on both p and n (D) positive **LG0143**
3. The number $\log_2 7$ is
 (A) an integer (B) a rational number
 (C) an irrational number (D) a prime number **LG0015**
4. If a, b, c are positive real numbers such that $a^{\log_3 7} = 27$; $b^{\log_7 11} = 49$ and $c^{\log_{11} 25} = \sqrt{11}$. The value of $\left(a^{(\log_3 7)^2} + b^{(\log_7 11)^2} + c^{(\log_{11} 25)^2}\right)$ equals
 (A) 489 (B) 469 (C) 464 (D) 400 **LG0074**
5. If $2^a = 3$ and $9^b = 4$ then value of (ab) is -
 (A) 1 (B) 2 (C) 3 (D) 4 **LG0001**
6. $\log_{10}(\log_2 3) + \log_{10}(\log_3 4) + \log_{10}(\log_4 5) + \dots + \log_{10}(\log_{1023} 1024)$ simplifies to
 (A) a composite (B) a prime number
 (C) rational which is not an integer (D) an integer **LG0144**
7. If $\log_2(x^2 + 1) + \log_{13}(x^2 + 1) = \log_2(x^2 + 1) \log_{13}(x^2 + 1)$, ($x \neq 0$), then $\log_7(x^2 + 24)$ is equal to
 (A) 1 (B) 2 (C) 3 (D) 4 **LG0006**
8. Given $\log_3 a = p = \log_b c$ and $\log_b 9 = \frac{2}{p^2}$. If $\log_9 \left(\frac{a^4 b^3}{c}\right) = \alpha p^3 + \beta p^2 + \gamma p + \delta$ ($\forall p \in \mathbb{R} - \{0\}$), then $(\alpha + \beta + \gamma + \delta)$ equals
 (A) 1 (B) 2 (C) 3 (D) 4 **LG0007**
9. The greatest value of $(4\log_{10} x - \log_x(.0001))$ for $0 < x < 1$ is -
 (A) 4 (B) -4 (C) 8 (D) -8 **LG0014**

10. If $60^a = 3$ and $60^b = 5$ then the value of $12^{\frac{1-a-b}{2(1-b)}}$ equals
 (A) 2 (B) 3 (C) $\sqrt{3}$ (D) $\sqrt{12}$ **LG0017**
11. Let ABC be a triangle right angled at C. The value of $\frac{\log_{b+c} a + \log_{c-b} a}{\log_{b+c} a \cdot \log_{c-b} a}$ ($b+c \neq 1, c-b \neq 1$) equals
 (A) 1 (B) 2 (C) 3 (D) $1/2$ **LG0018**
12. If α and β are the roots of the equation $(\log_2 x)^2 + 4(\log_2 x) - 1 = 0$ then the value of $\log_\beta \alpha + \log_\alpha \beta$ equals
 (A) 18 (B) -16 (C) 14 (D) -18 **LG0019**
13. If $\log_2 (4 + \log_3 (x)) = 3$, then sum of digits of x is -
 (A) 3 (B) 6 (C) 9 (D) 18 **LG0002**
14. Sum of all the solution(s) of the equation $\log_{10}(x) + \log_{10}(x+2) - \log_{10}(5x+4) = 0$ is -
 (A) -1 (B) 3 (C) 4 (D) 5 **LG0003**
15. The product of all the solutions of the equation $x^{1+\log_{10} x} = 100000x$ is -
 (A) 10 (B) 10^5 (C) 10^{-5} (D) 1 **LG0004**
16. If x_1 and x_2 are the roots of equation $e^{3/2} \cdot x^{2\log x} = x^4$, then the product of the roots of the equation is -
 (A) e^2 (B) e (C) $e^{3/2}$ (D) e^{-2} **LG0005**
17. If $\log_a (1 - \sqrt{1+x}) = \log_a (3 - \sqrt{1+x})$, then number of solutions of the equation is -
 (A) 0 (B) 1 (C) 2 (D) infinitely many **LG0008**
18. The number of solution(s) of $\sqrt{\log_3 (3x^2) \cdot \log_9 (81x)} = \log_9 x^3$ is -
 (A) 0 (B) 1 (C) 2 (D) 3 **LG0009**
19. If $x, y \in 2^n$ when $n \in \mathbb{I}$ and $1 + \log_x y = \log_2 y$, then the value of $(x+y)$ is -
 (A) 2 (B) 4 (C) 6 (D) 8 **LG0011**
20. The number of integral solutions of $|\log_5 x^2 - 4| = 2 + |\log_5 x - 3|$ is -
 (A) 1 (B) 2 (C) 3 (D) 0 **LG0016**

21. If $\log_{10} \left(\frac{1}{2^x + x - 1} \right) = x(\log_{10} 5 - 1)$, then $x =$
 (A) 4 (B) 3 (C) 2 (D) 1 LG0021
22. The number of solutions of the equation $\log_{x-3}(x^3 - 3x^2 - 4x + 8) = 3$ is equal to
 (A) 4 (B) 3 (C) 2 (D) 1 LG0022
23. Sum of the roots of the equation $9^{\log_3(\log_2 x)} = \log_2 x - (\log_2 x)^2 + 1$ is equal to
 (A) 2 (B) 4 (C) 6 (D) 8 LG0023
24. If 1, $\log_9(3^{1-x} + 2)$ and $\log_3(4 \cdot 3^x - 1)$ are in A.P. then x can be
 (A) $\log_4 3$ (B) $\log_3 4$ (C) $1 - \log_3 4$ (D) $\log_3 0.25$ LG0025
25. Number of real solutions of the equation $\sqrt{\log_{10}(-x)} = \log_{10} \sqrt{x^2}$ is :
 (A) zero (B) exactly 1 (C) exactly 2 (D) 4 LG0145
26. If x_1 & x_2 are the two values of x satisfying the equation $7^{2x^2} - 2(7^{x^2+x+12}) + 7^{2x+24} = 0$, then $(x_1 + x_2)$ equals -
 (A) 0 (B) 1 (C) -1 (D) 7 LG0010
27. If $\log_{0.3}(x-1) < \log_{0.09}(x-1)$, then x lies in the interval
 (A) $(2, \infty)$ (B) $(1, 2)$ (C) $(1, \infty)$ (D) none of these LG0020
28. If x satisfies the inequality $\log_{25} x^2 + (\log_5 x)^2 < 2$, then $x \in$
 (A) $\left(\frac{1}{25}, 5 \right)$ (B) $(1, 2)$ (C) $(4, 5)$ (D) $(0, 1)$ LG0024
29. If $x = \log_2 \left(\sqrt{56 + \sqrt{56 + \sqrt{56 + \sqrt{56 + \dots \infty}}} \right)$, then which of the following statements holds good ?
 (A) $x < 0$ (B) $0 < x < 2$ (C) $2 < x < 4$ (D) $3 < x < 4$ LG0013
30. If $n \in \mathbb{N}$ such that characteristic of n^2 to the base 8 is 2, then number of possible values of n is -
 (A) 14 (B) 15 (C) 448 (D) infinite LG0012

EXERCISE (O-2)

Multiple Correct Answer Type

1. If $\log_a x = b$ for permissible values of a and x then identify the statement(s) which can be correct ?
 (A) If a and b are two irrational numbers then x can be rational.
 (B) If a rational and b irrational then x can be rational.
 (C) If a irrational and b rational then x can be rational.
 (D) If a rational and b rational then x can be rational. LG0146
2. Which of the following statements is(are) correct ?
 (A) $7^{1/7} > (42)^{1/14} > 1$ (B) $\log_3(5) \log_7(9) \log_{11}(13) > -2$
 (C) $\sqrt{99+70\sqrt{2}} + \sqrt{99-70\sqrt{2}}$ is rational (D) $\frac{1}{\log_4 3} + \frac{1}{\log_7 3} > 3$ LG0027
3. Let $N = \frac{\log_2 24}{\log_{96} 2} - \frac{\log_2 192}{\log_{12} 2}$. Then N is : LG0147
 (A) a natural number (B) a prime number (C) a rational number (D) an integer
4. If $(\log_\beta \alpha)^2 + (\log_\alpha \beta)^2 = 79$, ($\alpha > 0$, $\beta > 0$, $\alpha \neq 1$, $\beta \neq 1$) then value of $(\log_\beta \alpha) + (\log_\alpha \beta)$ can be -
 (A) 7 (B) -9 (C) 9 (D) -7 LG0026
5. For $a > 0$, $\neq 1$ the roots of the equation $\log_{ax} a + \log_x a^2 + \log_{a^2x} a^3 = 0$ can be
 (A) $a^{-3/4}$ (B) $a^{-4/3}$ (C) $a^{-1/2}$ (D) none of these LG0028
6. The solution of the equation $5^{\log_a x} + 5x^{\log_a 5} = 3$, ($a > 0$, $a \neq 1$) is
 (A) $a^{-\log_5 2}$ (B) $a^{\log_5 2}$ (C) $2^{-\log_5 a}$ (D) $2^{\log_5 a}$ LG0029
7. The equation $x^{(3/4)(\log_2 x)^2 + \log_2 x - 5/4} = \sqrt{2}$ has
 (A) at least one real solution (B) exactly three real solutions
 (C) exactly one irrational solution (D) real + non real roots LG0031
8. The equation $\log_{x^2} 16 + \log_{2x} 64 = 3$ has :
 (A) one irrational solution (B) no prime solution
 (C) two real solutions (D) one integral solution LG0148

9. The equation $x^{\left[(\log_3 x)^2 - \frac{9}{2} \log_3 x + 5\right]} = 3\sqrt{3}$ has
 (A) exactly three real solution (B) at least one real solution
 (C) exactly one irrational solution (D) complex roots **LG0149**
10. Let a and b be real numbers greater than 1 for which there exists a positive real number c, different from 1, such that $2(\log_a c + \log_b c) = 9 \log_{ab} c$, then the possible value of $\log_a b$.
 (A) $1/2$ (B) 3 (C) $1/3$ (D) 2 **LG0071**
11. If x_1, x_2 are the value(s) of x satisfying the equation $\log_2^2(x-2) + \log_2(x-2) \log_2\left(\frac{3}{x}\right) - 2 \log_2^2\left(\frac{3}{x}\right) = 0$, then $x_1 + x_2$ is equal to
 (A) $x_1 + x_2 = 9$ (B) $x_1 x_2 = 18$ (C) $|x_1 - x_2| = 3$ (D) $x_1^2 + x_2^2 = 45$ **LG0040**
12. The solution set of the system of equations $\log_3 x + \log_3 y = 2 + \log_3 2$ and $\log_{27}(x+y) = \frac{2}{3}$ is :
 (A) {6, 3} (B) {3, 6} (C) {6, 12} (D) {12, 6} **LG0151**
13. If $2^{x+y} = 6^y$ and $3^{x-1} = 2^{y+1}$, then the value of $(\log 3 - \log 2)/(x - y)$ is
 (A) 1 (B) $\log_2 3 - \log_3 2$ (C) $\log(3/2)$ (D) $\log 3 - \log 2$ **LG0030**
14. The inequality $-1 \leq \left(\frac{1}{3}\right)^x < 2$ is satisfied by
 (A) $x \in [0, 1]$ (B) $x > -\log_3 2$ (C) $x < -\log_3 2$ (D) $x < -1$ **LG0032**
15. Which of the following statements are true
 (A) $\log_2 3 < \log_{12} 10$ (B) $\log_6 5 < \log_7 8$
 (C) $\log_3 26 < \log_2 9$ (D) $\log_{16} 15 > \log_{10} 11 > \log_7 6$ **LG0150**

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 and 2

Let $\log_2 5 = a$; $\log_5 3 = b$

1. The value of $\log_{50} 30$ is equal to

(A) $\frac{b+ab+1}{b+2}$ (B) $\frac{a+ab+1}{2a+1}$ (C) $\frac{a+ab+1}{a+2}$ (D) $\frac{a+ab+b}{2a+1}$

LG0033

2. Roots of the equation $\log_{10} 2 \cdot \log_{10} 5x^2 - (\log_{10} 5 + \log_{10} 15 \cdot \log_{10} 2)x + \log_{10} 15 = 0$ are

(A) **a, b** (B) $1+a, 1+b$ (C) $a-1, 1-b$ (D) $a+2; b+2$

LG0033

Paragraph for Question 3 and 4

Let $\log_5 N = I_1 + f_1$ and $\log_3 N = I_2 + f_2$, where I_1, I_2 are integers and $f_1, f_2 \in [0,1)$.

On the basis of above information, answer the following questions :

3. If $I_1 = 2$ and $I_2 = 3$, then maximum integral value of N is greater than or equal to -

(A) 79 (B) 80 (C) 81 (D) 82 LG0152

4. If $I_1 = 3$ and $I_2 = 4$, then number of possible integral values of N is less than or equal to -

(A) 119 (B) 116 (C) 117 (D) 118 LG0153

Paragraph for Question 5 and 6

If (x_1, y_1) and (x_2, y_2) are the solution of the system of equation.

$$\log_{225}(x) + \log_{64}(y) = 4$$

$$\log_x(225) - \log_y(64) = 1,$$

5. The value of $\log_{30}(x_1 y_1 x_2 y_2)$ is

LG0154

6. Number of digits in $(x_1 y_1 x_2 y_2)$ is

LG0097

Matrix Match Type

7. Match List-I with List-II and select the correct answer using the code given below the list.

List-I

List-II

(P) The value of x for which

(1) 2

$$2\ln(2^x - 5) = \ln 2 + \ln\left(2^x - \frac{7}{2}\right) \text{ is}$$

(2) 4

(Q) The sum of the real roots of the equation

(3) 3

$$x^{\log_{10}\left(\frac{5x}{2}\right)} = 10^{\log_{10} x} \text{ is}$$

(4) 5

(R) If $\alpha = 2^{\frac{(\log_3(\log_3 5))}{\log_3 2}}$ and $\beta^\alpha = 9$, then value

(5) $\frac{1}{2}$

$$\frac{1}{5}(\beta^{\alpha^2}) \text{ is equal to}$$

(S) $\log_{\sqrt{3}}\left(\sqrt{6 + \sqrt{6 + \sqrt{6 + \dots\infty}}}\right)$ is equal to

The correct option is

(A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 2; S \rightarrow 5$

(B) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 5$

(C) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

(D) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 4; S \rightarrow 1$

LG0155

8. Match List-I with List-II and select the correct answer using the code given below the list.

List-I

List-II

(P) Positive value of x satisfying the equation

(1) 5

$$(25)^{\frac{2x^2 - 5x - 9}{4}} = (\sqrt{11})^{3\log_{11} 5} \text{ is}$$

(Q) Sum of values of x which satisfy the equation

(2) 12

$$\sqrt{|x - 4|^{x-2}} = \sqrt[4]{|x - 4|^{x+2}} \text{ is}$$

(R) The expression $\sqrt{\log_{0.5}^2 32}$ has the value equal to

(3) 4

(S) The value of 3k, where

(4) 18

$$\log(\log 9) + \log(\log 49) = \log k + \log(\log 3) + \log(\log 7), \text{ is}$$

(A) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$

(B) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 2$

(C) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$

(D) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$

LG0156

9. In the following matrix match **Column-I** has some quantities and **Column-II** has some comments or other quantities

Match the each element in **Column-I** with corresponding element(s) in **Column-II**

Column-I	Column-II
(A) $\log_{\sin \pi/6} e$	(P) $x^{\log_x 2} ; x > 0 ; x \neq 1$
(B) 2	(Q) $2^{\log_5 7}$
(C) $\log_{\tan \pi/3} \pi$	(R) negative
(D) $7^{\log_5 2}$	(S) positive

LG0157

Column-I	Column-II
(A) Anti logarithm of $(0.\overline{6})$ to the base 27 has the value equal to	(p) 5
(B) Characteristic of the logarithm of 2008 to the base 2 is	
(C) The value of b satisfying the equation, $\log_e 2 \cdot \log_b 625 = \log_{10} 16 \cdot \log_e 10$ is	(q) 7
(D) Number of naughts after decimal before a significant figure comes in the number $\left(\frac{5}{6}\right)^{100}$, is	(r) 9 (s) 10

(Given $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$)

EXERCISE (O-4)

Numerical Grid Type

1. Let A denotes the value of $\log_{10} \left(\frac{ab + \sqrt{(ab)^2 - 4(a+b)}}{2} \right) + \log_{10} \left(\frac{ab - \sqrt{(ab)^2 - 4(a+b)}}{2} \right)$
when $a = 43$ and $b = 57$ and B denotes the value of the expression $(2^{\log_6 18}) \cdot (3^{\log_6 3})$.
Find the value of (A.B). LG0045
2. Let $y = \sqrt{\log_2 3 \cdot \log_2 12 \cdot \log_2 48 \cdot \log_2 192 + 16} - \log_2 12 \cdot \log_2 48 + 10$. Find $y \in \mathbb{N}$. LG0089
3. There exist positive integers A, B and C with no common factor greater than 1 such that $A \log_{200} 5 + B \log_{200} 2 = C$, then the value of $A + B + C$ is LG0043
4. If $a = \log_{12} 18$ & $b = \log_{24} 54$ then find the value of $ab + 5(a - b)$. LG0077
5. Number of solutions of $\log_{(1+x)} \left(\frac{1-x}{1+x} \right) - 2 \log_{(1-x)} (1+x) = 0$ is LG0038
6. The minimum possible real x which satisfy the equation, $2 \log_2 \log_2 x + \log_{1/2} \log_2 (2\sqrt{2}x) = 1$. LG0044
7. Find the square of the sum of the solution of the equation
 $\log_3 x \cdot \log_4 x \cdot \log_5 x = \log_3 x \cdot \log_4 x + \log_4 x \cdot \log_5 x + \log_5 x \cdot \log_3 x$. LG0048
8. If 'x' and 'y' are real numbers such that $2 \log(2y - 3x) = \log x + \log y$, find $\frac{y+x}{y-x}$. LG0075
9. If $4^A + 9^B = 10^C$, where $A = \log_{16} 4$, $B = \log_3 9$ & $C = \log_8 3$, then find x. LG0158
10. Find the number of positive integers not satisfying the inequality $\log_7 (4^x - 2 \cdot 2^x + 17) > 5$. LG0159

EXERCISE (JA)

1. Number of solutions of $\log_4(x-1) = \log_2(x-3)$ is [JEE 2001 (Screening)]
 (A) 3 (B) 1 (C) 2 (D) 0

LG0106

2. Let (x_0, y_0) be the solution of the following equations [JEE 2011, 3 (-1)]

$$(2x)^{\ln 2} = (3y)^{\ln 3}$$

$$3^{\ln x} = 2^{\ln y}$$

Then x_0 is

- (A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) 6

LG0107

3. The value of $6 + \log_{\frac{3}{2}} \left(\frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}}} \sqrt{4 - \frac{1}{3\sqrt{2}}} \sqrt{4 - \frac{1}{3\sqrt{2}}} \dots \right)$ is [JEE 2012, 4M]

LG0108

4. If $3^x = 4^{x-1}$, then $x =$ [JEE-Advanced 2013, 4, (-1)]

- (A) $\frac{2\log_3 2}{2\log_3 2 - 1}$ (B) $\frac{2}{2 - \log_2 3}$ (C) $\frac{1}{1 - \log_4 3}$ (D) $\frac{2\log_2 3}{2\log_2 3 - 1}$

LG0109

5. The value of $((\log_2 9)^2)^{\frac{1}{\log_2(\log_2 9)}} \times (\sqrt{7})^{\frac{1}{\log_4 7}}$ is _____ [JEE(Advanced)-2018]

LG0110

6. The product of all positive real values of x satisfying the equation $x^{(16(\log_5 x)^3 - 68\log_5 x)} = 5^{-16}$ is _____. [JEE(Advanced)-2022]

LG0160

ANSWER KEY

Do yourself - 1

1. (a) $\log_3 81 = 4$ (b) $\log_{10}(0.001) = -3$ (c) $\log_{128} 2 = 1/7$
2. (a) $32 = 2^5$ (b) $4 = (\sqrt{2})^4$ (c) $0.01 = 10^{-2}$
3. 6
4. (i) 0 (ii) 1 (iii) 2 (iv) 3 (v) -1
(vi) -5 (vii) -4 (viii) 1/2 (ix) 1 (x) 3/2
(xi) -1/5 (xii) -3/7
5. (i) 0 (ii) 1 (iii) 3 (iv) -4 (v) -1/2
(vi) 1/2 (vii) -3/2 (viii) 9/4
6. (i) 0 (ii) 1 (iii) 2 (iv) 4 (v) -1
(vi) 1/2 (vii) -3/2 (viii) 7/2 (ix) 2/7
7. (i) 0 (ii) 1 (iii) 2 (iv) -1 (v) -2
(vi) -4 (vii) -1/3 (viii) 1/7 (ix) -5/2 (x) 9/4
8. (i) 0 (ii) 1 (iii) 2 (iv) 4 (v) -1
(vi) -2 (vii) -1/2 (viii) 1/4 (ix) 1/2 (x) 1/3
(xi) 1/3
9. (i) 4 (ii) 2 (iii) $a > 0, a \neq 1$ (iv) -5, 2 (v) -2, 2
(vi) -3, 3 (vii) 9 (viii) 1/81 (ix) 1 (x) 2
(xi) $\sqrt{2}, -\sqrt{2}$
10. (i) $-\frac{1}{2}$ (ii) 2 (iii) 1 (iv) 1 (v) -1

Do yourself - 2

1. (a) 1 (b) $\frac{1}{5}$ 2. 3
4. (i) 2 (ii) 3 (iii) 1 (iv) 4

Do yourself - 3

1. (i) 3 (ii) 5/6 (iii) 0 (iv) 2 (v) (A)
(vi) (C)
2. (i) 3 (ii) 9 (iii) 9 (vi) 2
3. $2s + 10s^2 - 3(s^3 + 1)$ 5. $\log_b N$ 6. 4 7. 9
8. 1 9. 6 10. $\frac{25}{2}$ 11. 1/6 12. 34
13. 32

Do yourself - 4

1. (i) $\sqrt{2}, -\sqrt{2}$ (ii) 1 (iii) 1 (iv) 0 (v) 1, -1
2. (i) $x = \pm 2$ (ii) $x = \pm \frac{1}{2}$ (iii) $x = \frac{3}{2}$ (iv) $x = 4$
3. $\log_2 3$ 4. $\frac{1}{8}, 2$ 5. $9, \frac{1}{9}$ 6. 2, 8 7. 1, 0
8. ϕ 9. $100, \frac{1}{100}$ 10. $2^{-\log_3 a}$ 11. ϕ 12. 2
13. $x = 8, \frac{1}{8}$ 14. $\{1 + \sqrt{3}\}$ 15. $\{3\}$ 16. $\{4\}$ 17. $\{2\}$
18. $\{0\}$ 19. $\{-5\}$ 20. $\{3/2, 10\}$ 21. $\{10^{-1}, 10\}$ 22. $\{\sqrt{5}, 5\}$
23. $\{-(3 - \sqrt{3})/3, 8\}$

Do yourself-5

1. $x \in (1, 8)$ 2. $x \in (-\infty, 1/2)$ 3. $(-1, 0) \cup (2, 3)$
4. $\frac{1}{\sqrt{10^9}} \leq x \leq 10$ 5. $-1 < x < \frac{1 - \sqrt{5}}{2}$ or $\frac{1 + \sqrt{5}}{2} < x < 2$
6. $2^{-\sqrt{2}} < x < 2^{-1}$; $1 < x < 2^{\sqrt{2}}$ 7. $(-\infty, -2.5) \cup (0, \infty)$
8. $\frac{1}{2} < x < 1$ 9. $(-2, -1) \cup (-1, 0) \cup (0, 1) \cup (2, \infty)$ 10. $\left(\frac{1}{2}, 1\right)$

Do yourself - 6

1. $x = 3$ 2. $x = -2$ 3. $x = 3$ 4. $x = 0$
5. $x = \frac{-7}{2}, 2$ 6. $x = \frac{1}{5}, 6$ 7. $x = 1$ 8. $x \in (-1, 1)$
9. $\left(0, \frac{1}{2}\right) \cup (32, \infty)$ 10. $\left(-1, -\frac{2}{\sqrt{5}}\right) \cup \left(\frac{2}{\sqrt{5}}, 1\right)$ 11. $(-\infty, \log_2 3)$ 12. $(-1, 0) \cup (0, 1)$

Do yourself - 7

1. -7.3314 2. 26 3. 155 4. 23040 5. 1
6. 5 7. 59

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	A	C	B	A	D	B	C	D	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	D	C	C	D	A	A	B	D	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	D	A	C	C	B	A	A	C	B

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C,D	A,B,D	A,B,C,D	B,C	B,C	A,C	A,B,C,D	A,B,C,D	A,B,C,D	A,D
Que.	11	12	13	14	15					
Ans.	A,B,C,D	A,B	C,D	A,B	B,C					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8		
Ans.	B	B	A,B	A,D	12.00	18.00	D	A		
Q.9	A	B	C	D	Q.10	A	B	C	D	
	R	P,S	S	Q,S		r	s	p	q	

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	12	6	6	1	0	8.00	3721.00	2.60	10.00	2.00

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6				
Ans.	B	C	4	A,B,C	8	1				

Important Notes

